

ISystem
ECS & _ecs
+ ISystem(ECS &ecs)
+ virtual ~ISystem() =default
+ virtual void update (double dt)=0
+ ISystem(ECS &ecs)
+ virtual ~ISystem() =default
+ virtual void update (double dt)=0



ParticleSystem
- std::vector< Particle_t > _pool - size_t _poolSize - sf::CircleShape _circleShape - std::mt19937 _rng - std::uniform_real_distribution< float > _dist
+ ParticleSystem(ECS &ecs, size_t poolSize=2000) + void update(double dt) override + void render(sf::RenderWindow &window) + void emit(ParticleType type, float x, float y, int count=20) + void emitCustom(float x, float y, int count, float speed, float spread, float lifetime, sf::Color startColor, sf::Color endColor, float startSize, float endSize, float gravity=0.f) + void emitExplosion(float x, float y, int count=30) + void emitSparks(float x, float y, int count=15) + void emitSmoke(float x, float y, int count=10) + void emitDebris(float x, float y, int count=8) + void emitTrail(float x, float y, float direction Deg, int count=3) - Particle_t * getParticle() - void updateParticle(Particle_t &p, float dt) - void spawnParticle(Particle_t &p, float x, float y, const ParticleEmitter_t &emitter) - sf::Color lerpColor(const sf::Color &a, const sf::Color &b, float t) - float randomFloat(float min, float max)